

Calc-Silicate Metamorphism & Sillimanite-Equivalent High-Grade Facies

Authoritative Reference: John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., Chapter 29: Metamorphism of Calcareous & Ultramafic Rocks)

Comprehensive Chapter Masterclass: Regional Metamorphism of Calc-Silicate Rocks

Marly Protoliths, Mixed-Volatile Phase Equilibria & Correlation with Barrovian Pelitic Zones

1. Calc-Silicate Rocks & Cross-Facies Correlation

Calc-silicate rocks are metamorphosed marls or calcareous mudstones/sandstones containing significant amounts of both carbonate components (CaCO_3 , $\text{CaMg}(\text{CO}_3)_2$) and silicate detritus (quartz, feldspar, illite/kaolinite clays providing Al, Si, and K). Because they contain both calcium and aluminum, calc-silicates develop distinctive Ca-Al-Mg silicates (zoisite/clinozoisite, grossular, anorthite, diopside, wollastonite) that are exceptionally sensitive to metamorphic grade.

In a landmark classic study, **W. Q. Kennedy (1949)** established the direct correlation between the progressive mineral zones in calc-silicate rocks and the Barrovian zones in associated pelitic schists of the Scottish Highlands:

- **Biotite Zone Equivalent:** *Zoisite + Calcite + Biotite + Albite + Quartz*
- **Garnet Zone Equivalent:** *Clinozoisite + Calcite + Hornblende + Oligoclase*
- **Staurolite / Kyanite Zone Equivalent:** *Diopside + Anorthite + Calcite ± Hornblende*
- **Sillimanite Zone Equivalent (Peak Grade):** **Wollastonite + Grossular** (+ Diopside + Anorthite)

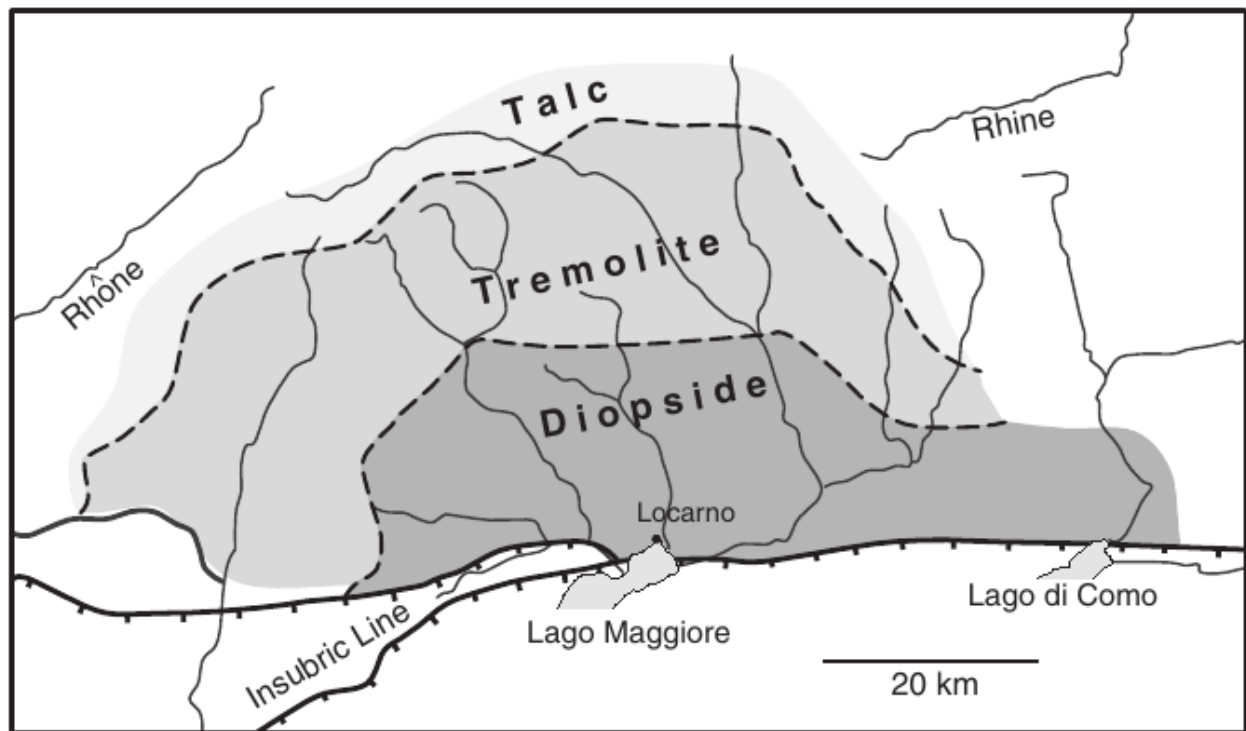


FIGURE 5 Metamorphic zones developed in regionally metamorphosed dolomitic rocks of the Lepontine Alps, along the Swiss–Italian border. After Trommsdorff (1966, 1972).

field gradient. The increase in pressure has two principal effects. Because devolatilization reactions at crustal pressures have a characteristic curved shape on a P - T phase diagram in which reaction temperature increases with increasing pressure, the reactions in Figure 6 are at higher temperature than in Figure 3. Second, if we compare invariant points A, B, and C between the two diagrams, we realize that the talc field is much smaller and the diopside field is much larger at higher pressure.

The reactions involved are the same as at low pressure, and rather than reiterate all the scenarios of the preced-

Figure 29-5 from John D. Winter (p. 677): Metamorphic zones developed in regionally metamorphosed dolomitic rocks of the Central Alps (after Trommsdorff, 1966). Note the progressive sequence: Tremolite → Diopside → Forsterite → Wollastonite + Grossular.

largest range of fluid compositions ($0.1 < X_{\text{CO}_2} < 0.87$), *tremolite* will be the first new metamorphic mineral to form [from dolomite + quartz via Reaction (5)]. From this point the sequence of reactions for open-system behavior is the same as for contact metamorphism. Closed-system behavior is highly favored at depths below 10 km, however, because the marbles will be compressed and permeability will be limited. Most reactions may thus be expected to internally buffer the fluids.

As an example of a typical situation, consider a path rising in temperature along the dashed curve at $X_{\text{CO}_2} = 0.5$ in Figure 6. When **tremolite** begins to form, the fluid will probably be buffered along the curve for Reaction (5) (dashed curve). The fluid composition becomes enriched in CO_2 as the reaction progresses, consuming H_2O and liberating CO_2 . If quartz is present in minor amounts, it may be consumed before the system reaches invariant point B, in which case the system would progress vertically from the curve at whatever temperature quartz disappears. If the example of Spear (1993) discussed above is correct, however, it is more likely that only a small degree of reaction progress is necessary for the fluid to be buffered to point B. At this point **diopside** will form along with dolomite, tremolite, quartz, and calcite in an invariant assemblage.

Which curve is followed from the invariant point again depends upon the relative proportions of the reacting phases (i.e., on the bulk composition). If tremolite and calcite are lost, the system may proceed along the $\text{Dol} + \text{Qtz} = \text{Di} + \text{CO}_2$ curve (Reaction 8) toward $X_{\text{CO}_2} = 1.0$. It is far more likely that quartz will be consumed first, however, and the system will follow the curve for Reaction (7), consuming tremolite and calcite to produce diopside + dolomite.

Figure 29-6 from John D. Winter (p. 677): $T-X_{\text{CO}_2}$ phase diagram for regionally metamorphosed calcareous rocks calculated at $P = 0.5 \text{ GPa}$ (5 kbar). The upper thermal boundary represents the entry into the **Wollastonite + Grossular** field, directly matching upper amphibolite to granulite conditions.

2. Petrologic Reactions Generating Wollastonite + Grossular

Under high amphibolite-to-granulite facies conditions equivalent to the pelitic **sillimanite zone** ($T > 650^\circ\text{C}$), two quintessential decarbonation reactions occur:

Reaction 1: The Wollastonite-Forming Decarbonation Reaction

Calcite reacts with excess quartz to eliminate the Cal-Qtz tie-line and produce wollastonite:



In regional terranes buffered to moderate-to-high X_{CO_2} , this reaction requires high temperatures ($> 650^\circ\text{C}$ – 720°C), precisely matching the sillimanite zone.

Reaction 2: The Grossular-Forming Decarbonation Reaction

Calcic plagioclase (anorthite) reacts with remaining calcite and quartz to produce calcic grossular garnet:



Alternatively, at high pressures, zoisite breaks down: $2 \text{ Zoisite} + \text{Calcite} + \text{Quartz} = 3 \text{ Grossular} + \text{CO}_2 + \text{H}_2\text{O}$.

Because both **wollastonite** and **grossular** require the complete breakdown of calcite with quartz and anorthite under peak thermal conditions, the assemblage **Wollastonite + Grossular** serves as the definitive high-grade calc-silicate marker equivalent to the sillimanite zone of pelites.



Official Exam Question Statement & In-Depth Critical Analysis

Directorate General of Mines Safety (DGMS) 2019 · Junior Scientific Officer Exam · Question 11

Question Statement

Q.11: In metamorphism of calc-silicate rocks, which assemblage can be considered equivalent to the sillimanite zone of pelitic rocks?

(a) Anorthite–pyroxene

(b) Zoisite–calcite–biotite

(c) Diopside–anorthite

(d) Wollastonite–grossular

Correct Answer: (d) Wollastonite–grossular



The high-temperature decarbonation assemblage of wollastonite + grossular defines the peak calc-silicate metamorphic grade, directly equivalent to the sillimanite zone in pelites.

Exhaustive Option-by-Option Critical Analysis

(d) Wollastonite–grossular

CORRECT

Why it is uniquely correct: In metamorphic petrology and regional metamorphic zonation (Winter, 2010, Chapter 29; Kennedy, 1949; Bucher & Grapes, 2011), the stable coexistence of **wollastonite** (CaSiO_3) and **grossular** ($\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$) requires high temperatures ($> 650^\circ\text{C}$ – 750°C) to drive the univariant decarbonation reactions $\text{Ca} + \text{Qtz} = \text{Wo} + \text{CO}_2$ and $\text{An} + 2\text{Ca} + \text{Qtz} = \text{Grs} + 2\text{CO}_2$. This peak assemblage represents the highest calc-silicate zone in regional metamorphic terranes and contact aureoles, directly equivalent to the upper amphibolite-to-granulite facies (the **sillimanite zone** of pelitic rocks).

(a) Anorthite–pyroxene

INCORRECT

Why it is incorrect: While anorthite and pyroxene (diopside) can occur across a broad temperature range in calc-silicate rocks, "anorthite–pyroxene" is more typically the characteristic signature of *mafic granulites*. In calc-silicates, anorthite and diopside form at intermediate grades (staurolite/kyanite zone); at the sillimanite grade, anorthite breaks down with calcite to produce grossular!

(b) Zoisite–calcite–biotite

INCORRECT

Why it is incorrect: In Kennedy's (1949) classic zonal correlation, the assemblage *zoisite* + *calcite* + *biotite* + *albite* is explicitly the diagnostic marker of the **biotite zone** (low-grade greenschist facies, $T < 400^{\circ}\text{C}$ – 450°C). Zoisite is a hydrous epidote-group mineral that is completely unstable at the high temperatures of the sillimanite zone.

(c) Diopside–anorthite

INCORRECT

Why it is incorrect: The *diopside*–*anorthite* assemblage corresponds to the **middle amphibolite facies (kyanite zone equivalent)**. Under progressive metamorphism, diopside and anorthite form before wollastonite and grossular appear. Only when temperatures rise further into the sillimanite zone does quartz react with calcite to generate wollastonite and anorthite react with calcite to yield grossular.



Synthesis Reference Tables, Diagnostic Criteria & High-Yield Practice

Comprehensive Zonal Equivalence Across Protoliths & 8 Exam-Grade Multiple-Choice Questions

Cross-Facies Zonal Equivalence: Pelites vs. Calc-Silicates vs. Metabasites

Barrovian Pelite Zone	Approx. T (°C)	Calc-Silicate Assemblage (Kennedy, 1949 / Winter, 2010)	Metabasite (Mafic) Facies
Chlorite Zone	< 350°C	Calcite + Quartz + Chlorite + Albite + Illite	Lower Greenschist (Actinolite + Chlorite + Albite)
Biotite Zone	350 – 450°C	Zoisite + Calcite + Biotite + Albite + Quartz	Middle Greenschist Facies
Garnet Zone	450 – 520°C	Clinozoisite + Calcite + Hornblende + Oligoclase	Epidote-Amphibolite Facies
Staurolite / Kyanite Zone	520 – 650°C	Diopside + Anorthite + Calcite ± Hornblende	Amphibolite Facies (Hornblende + Plagioclase)
Sillimanite Zone	> 650 – 750°C	Wollastonite + Grossular (+ Diopside + Anorthite)	Upper Amphibolite to Granulite Facies

8 High-Yield Practice Questions on Calc-Silicates & Metamorphic Zonation

1. In Kennedy's (1949) zonal correlation of metamorphic rocks in the Scottish Highlands, which calc-silicate assemblage corresponds to the biotite zone of pelites?

- (a) Wollastonite + Grossular (b) Zoisite + Calcite + Biotite (c) Diopside + Forsterite (d) Periclase + Brucite

Answer: (b) Zoisite + Calcite + Biotite — Low-grade marly rocks develop zoisite and biotite in equilibrium with calcite, matching the pelitic biotite isograd.

2. What is the fundamental decarbonation reaction that introduces wollastonite into high-grade calc-silicates?

- (a) Calcite + Quartz = Wollastonite + CO₂
(b) Dolomite + Quartz = Diopside + 2 CO₂

- (c) Tremolite + Calcite = Diopside + CO₂ + H₂O
 (d) Grossular + Quartz = Anorthite + Wollastonite

Answer: (a) Calcite + Quartz = Wollastonite + CO₂ — The classic univariant decarbonation equilibrium.

3. The presence of grossular garnet in calc-silicate rocks requires the presence of which element in the sedimentary protolith?

- (a) Titanium (b) Aluminum (from detrital clay/shale) (c) Zirconium (d) Phosphorus

Answer: (b) Aluminum — Pure carbonate lacks Al; grossular ($\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$) requires aluminum supplied by detrital clay minerals.

4. In contact metamorphic aureoles of limestone (e.g., Crestmore, California), the innermost, highest-temperature zone adjacent to the pluton is typically the:

- (a) Forsterite zone (b) Monticellite zone (c) Grossular–Wollastonite–Diopside zone (d) Chlorite zone

Answer: (c) Grossular–Wollastonite–Diopside zone — The innermost zone represents peak temperatures where silica metasomatism and intense decarbonation yield grossular, wollastonite, and diopside.

5. On an isobaric $T-X_{\text{CO}_2}$ diagram, why do devolatilization reaction curves that liberate both H₂O and CO₂ (e.g., tremolite formation) exhibit a parabolic maximum?

- (a) Due to changes in gravity
 (b) Because the reaction produces mixed volatiles, reaching maximum thermal stability at a specific fluid composition matching the stoichiometric product ratio ($X_{\text{CO}_2} = n_{\text{CO}_2} / (n_{\text{CO}_2} + n_{\text{H}_2\text{O}})$)
 (c) Because calcite melts at low temperatures
 (d) Because quartz dissolves into the gas phase

Answer: (b) — Reactions producing both H₂O and CO₂ reach their peak temperature on a $T-X_{\text{CO}_2}$ diagram where the fluid composition equals the ratio in which the gases are evolved.

6. Which calc-silicate mineral is an epidote-group sorosilicate with formula $\text{Ca}_2\text{Al}_3\text{Si}_3\text{O}_{12}(\text{OH})$?

- (a) Zoisite/Clinzoisite (b) Wollastonite (c) Diopside (d) Forsterite

Answer: (a) Zoisite/Clinzoisite — Zoisite (orthorhombic) and clinzoisite (monoclinic) are hydrated Ca-Al silicates characteristic of low-to-medium grade calc-silicates.

7. In high-grade regional calc-silicate rocks, scapolite commonly forms by replacement of:

(a) Plagioclase feldspar in the presence of CO_2 - and Cl -rich fluids (b) Olivine (c) Biotite (d) Pyrite

Answer: (a) — Scapolite (meionite/marialite solid solution) is an aluminosilicate containing volatile anions (CO_3^{2-} , Cl^- , SO_4^{2-}) that replaces plagioclase at high metamorphic temperatures.

8. The transition from the diopside–anorthite zone to the wollastonite–grossular zone in calc-silicate rocks coincides with which pelitic reaction?

- (a) Chlorite = Biotite
- (b) Muscovite + Quartz = Sillimanite + K-feldspar + H_2O (or melt)
- (c) Smectite = Illite
- (d) Kaolinite = Pyrophyllite

Answer: (b) — The second sillimanite isograd (muscovite breakdown to sillimanite + Kfs) occurs at roughly $650^\circ\text{--}700^\circ\text{C}$, matching the formation of wollastonite and grossular in calc-silicates.